

Superstore Sales & Profitability Analysis

Data Storytelling & Statistical Validation

Analyst

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Programme

Data Analytics Internship Project @ApexPlanet



Business Objective

This analysis explores the Superstore dataset to uncover actionable insights across sales, profitability, regions, and customer segments — driving smarter strategic decisions.



Analyse Trends

Examine sales and profit trends over time to identify growth patterns and seasonal fluctuations.



Identify Performance

Pinpoint high-performing and low-performing categories, sub-categories, and products.



Regional & Segment Insights

Understand how different regions and customer segments contribute to overall revenue.



Generate Recommendations

Deliver data-backed business recommendations validated through statistical analysis.



Dataset Overview & Tools

Dataset Summary

The Superstore dataset contains **9,994 transaction records** across **21 features**, capturing sales activity across the United States. Key columns analysed include:

Sales

Profit

Category

Region

Segment

Product Name

Tools & Technologies



Python



Pandas



Jupyter Notebook



Power BI / Excel

Key Performance Indicators

A snapshot of overall business performance derived from the Superstore dataset.

\$2.3M

Total Sales

Across all regions and segments

\$286K

Total Profit

Net profit across all transactions

12.5%

Profit Margin

Overall average margin

#1

Top Category

Technology

#1

Top Region

West

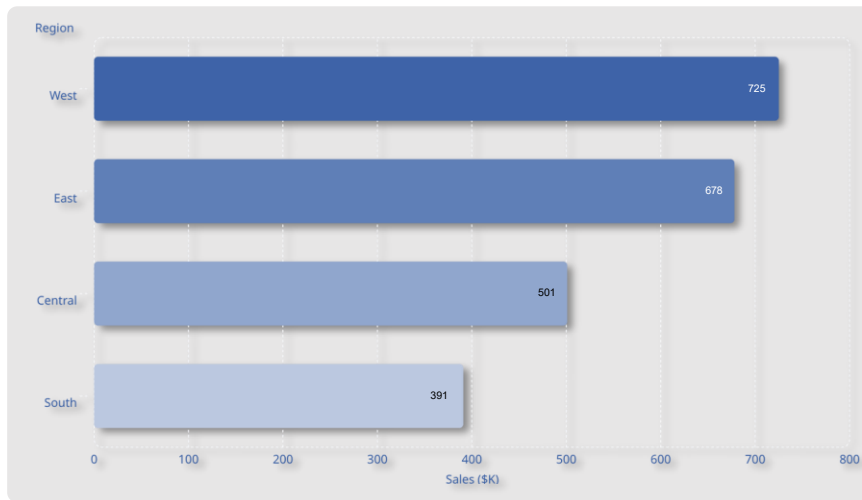
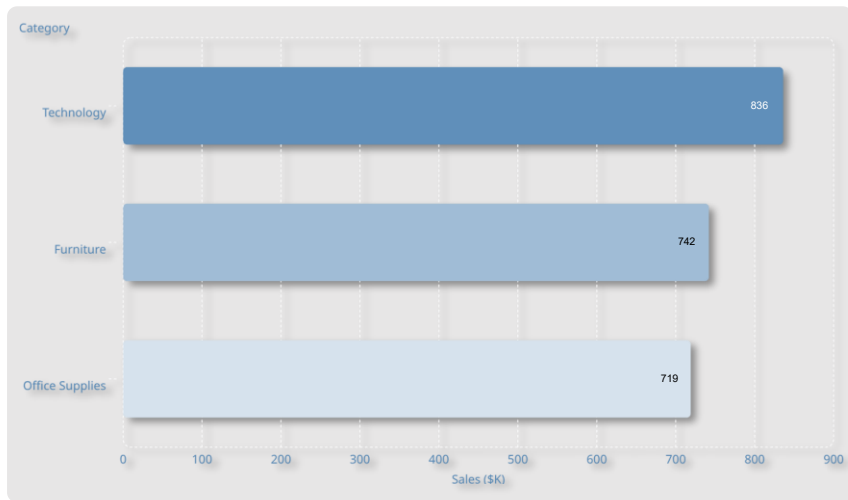
#1

Top Segment

Consumer

Sales Analysis

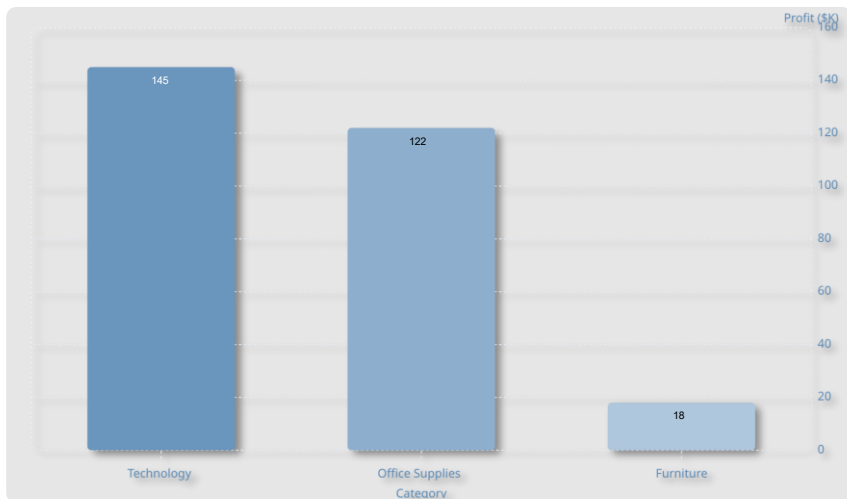
Technology leads all categories in revenue, while the West region drives the highest geographical contribution to total sales.



✔ **Key Insight:** Technology accounts for ~36% of total sales. The West region alone contributes nearly 32% of overall revenue — making it the most strategically valuable geography.

Profit Analysis

Whilst Technology generates both high sales and strong profits, Furniture trails significantly — highlighting a margin problem that warrants strategic intervention.



Sub-Category Profitability Highlights



Copiers

Highest sub-category profit margin



Phones

Strong sales with healthy margin



Tables

Negative profit — high discount pressure

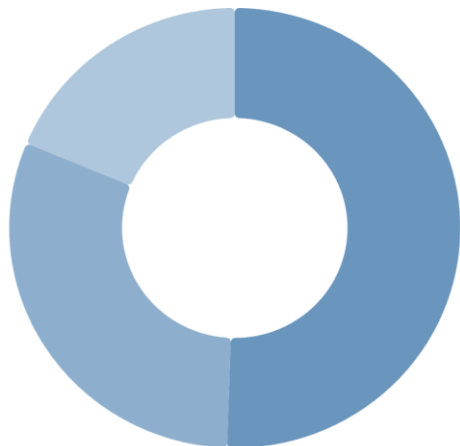


Bookcases

Low profitability despite decent volume

Customer & Product Insights

Segment Contribution to Sales



Consumer: 50.54%



Corporate: 30.74%



Home Office: 18.72%

Top-Performing Products

These sub-categories consistently rank highest in both revenue and volume across all regions:

01

Phones

Highest-grossing sub-category in Technology

02

Chairs

Top performer within the Furniture category

03

Storage

Consistent volume driver in Office Supplies

Interactive Dashboard Showcase

*An interactive dashboard was built in **Power BI / Excel** to deliver real-time business insights with dynamic filtering and drill-down capability.*

Dynamic Filters

Slice by region, segment, category, and date range instantly

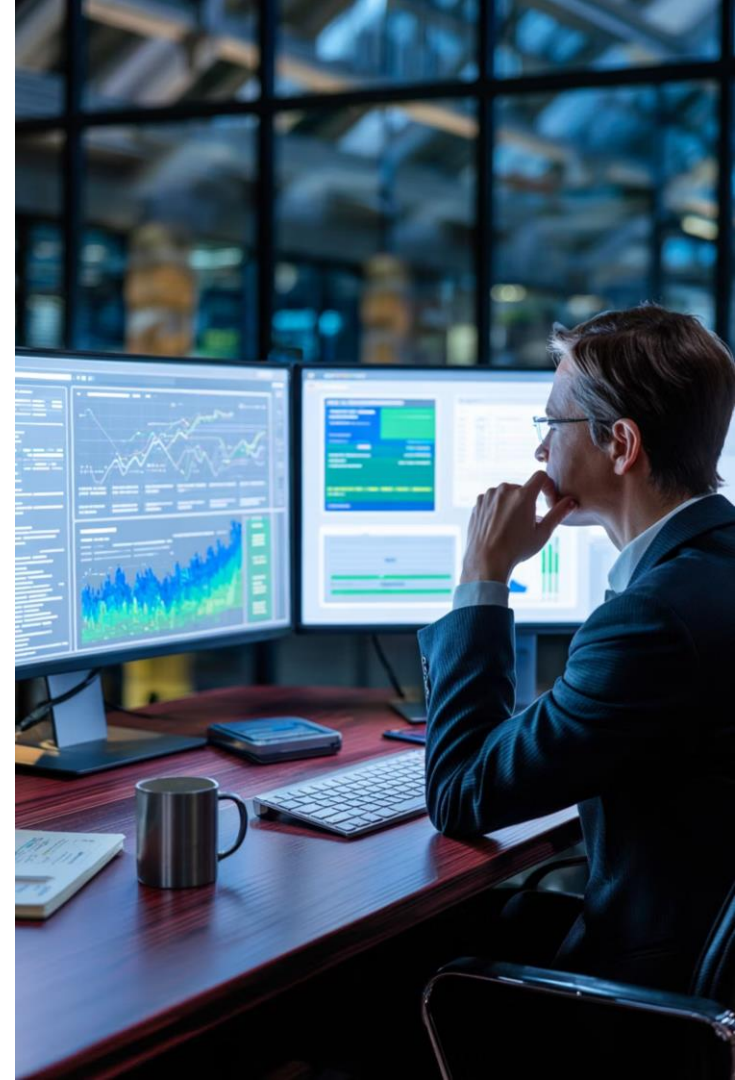


Live KPI Cards

Total Sales, Profit, and Margin update in real time

Visual Charts

Bar, donut, and map charts for rapid decision-making



Dashboard Screenshot

Total Orders

10K

Total Orders

Total Sales

\$2.30M

Total Sales

Total Profit

\$286.40K

Total Profit

Profit Margin

12.47%

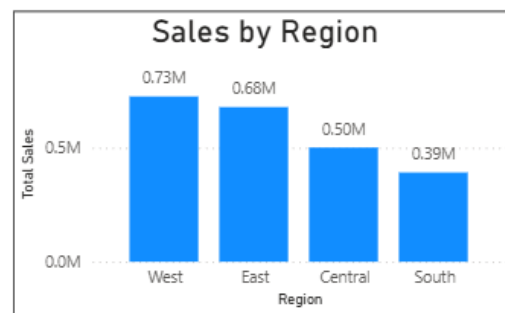
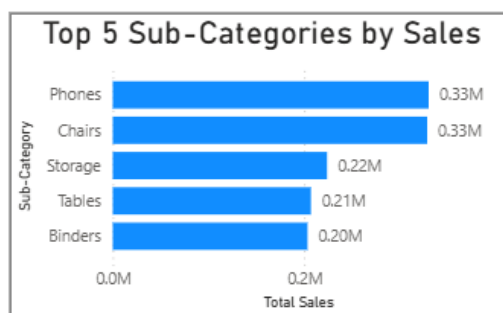
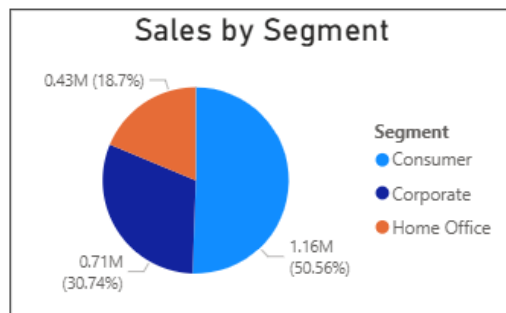
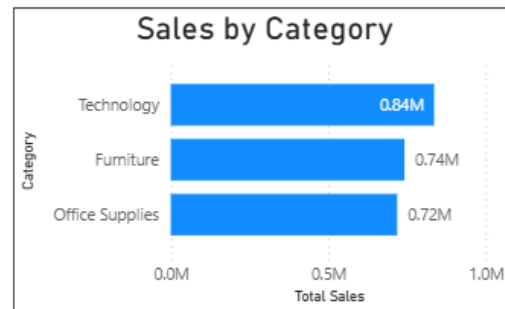
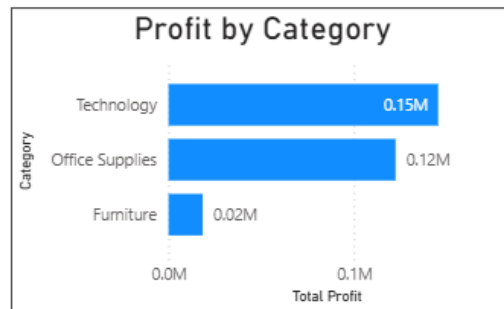
Profit Margin

AOV

\$229.86

AOV

Region	Segments	Category
Central	Consumer	Furniture
East	Corporate	Office Sup...
South	Home Offi...	Technology
West		



Statistical Validation & Hypothesis Testing

Business Question: Is the Technology category significantly more profitable than Furniture?

H₀: No difference in profit (Tech vs. Furniture)

Independent T-Test

p-value < 0.05

Reject H₀

**Conclude H₁:
Technology is
more profitable**

H₀ — Null Hypothesis

No significant difference in average profit between Technology and Furniture

H₁ — Alternative Hypothesis

Technology category has a statistically higher average profit

Method: Independent T-Test

p-value < 0.05 — Null hypothesis rejected at 95% confidence level



Conclusion: Technology is statistically and significantly more profitable than Furniture. Data validates the strategic case for increased investment.

Recommendations & Conclusion

1

Invest in Technology

Expand product range and marketing spend in the highest-margin category

2

Fix Furniture Margins

Revise discount strategies and pricing models for Tables and Bookcases

3

Double Down on the West

Prioritise marketing and logistics investment in the top-performing region

4

Strengthen Consumer Loyalty

Develop loyalty programmes targeting the Consumer segment — the largest revenue driver

 **Final Conclusion:** Technology is the star performer; Furniture demands profitability optimisation. Statistical testing validates these insights — empowering confident, data-driven strategic decisions.